

A Pattern Based Approach to Human Motion Control

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Abstract. Most researches on motion control are attempting to control motion from the outside. But as Bernstein pointed out in the case of human motion the number of the degrees of freedom is tremendously large, so it is extremely difficult to control motion from the outside. However, if we note how our bodies contribute to our cognition, there are approaches from the other way, i.e., from the inside. Gallwey pointed out in his book “The Inner Game of Tennis” what an important role our bodies play in tennis. Our muscles are different from person to person, so to win a game, there is no explicit way, but we must fully utilize our embodied cognition. It is his message. This paper describes a pattern-based approach to motion control, which is based on our capability of embodied cognition.

Keywords. Motion Control, Skill Transfer, Pattern Based Approach, Mahalanobis Taguchi System, Embodied Cognition

Introduction

Polanyi [1] pointed out there are two kinds of knowledge; explicit and tacit. Motion Control is a typical tacit knowledge. Although there are many machines which control motion, how we can control human motion is still not clear. Let us take a bicycle riding, for example. We learn to ride a bicycle by trial and error by ourselves.

Bernstein used cyclogram and pursued a human motion of hammering down. Near the object, his motion trajectories are almost identical, but far from it, his motion varies widely from time to time (Figure 1).

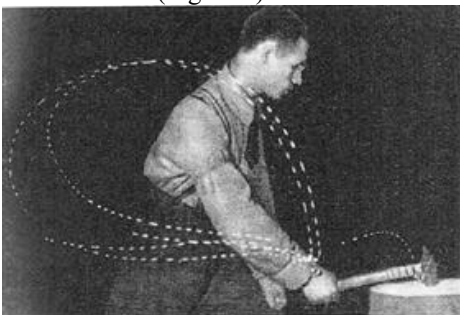


Figure 1. Cyclogram of hammering.

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Machines reproduce our motion near the goal. So it can be controlled rationally or explicitly. This is possible, because the environment does not change. When the environment changes, adaptive control may provide a solution. But even in such a case, there are objects that do not change with time. But in bicycle riding, situations are totally different. Everything changes with time.

Bernstein pointed out the difficulty of human motion control is due to its large degrees of freedom. Indeed it is, if we consider bicycle riding.

But the importance of human motion control is increasing. It is not only to transfer skills, but more to activate our cognitive capabilities [2].

It is pointed out in this paper how important human motion is and a new approach which is pattern-based, and is based on our intrinsic cognitive capabilities is proposed to control it.

1. Human motion control: state of the art

Human motion control has been attracting wide attention, because it relates to many different fields, which include sports, assistive technology, etc. Another important point we have to remember is even if we control our machines, we need to control our motions, because most machines are not fully automated.

Most human motion control studies use motion capture or EMG (Electromyography). In motion capture studies, most of them observe the movements of our joints. They reproduce our successful movements, but they are insufficient for activating or developing our cognitive capabilities. There are many unsuccessful data, but they do not provide us with any clue to how we can control ourselves to move successfully. EMG is used primarily for observing the health of muscles.

What is becoming increasingly important today is fast adaptability. The environments and situations change very frequently and extensively. So we have to be adaptable to cope with such changes. Human motion control is no exception. Yesterday, how we can transfer our skills was very important, but the skills needed today have changed. It becomes more important how adaptable we can be in controlling our motions. In this sense, the process of motion learning becomes more important. In other words, adaptable motion and bicycle riding have much in common.

To make it clear how we can tackle this problem, let us look back and understand how we expanded our controllability in engineering.

2. The world is changing

Yesterday, our world was closed with definite boundaries. In a closed world, we can apply rational approaches. Changes were not frequent and even if there were, they were gentle. So they were mathematically differentiable and we could predict the future.

But today, our world is expanding very rapidly and extensively. It is an open world without boundaries. Changes are frequent and extensive and further they are angular. Therefore, they are not differentiable so that we cannot predict the future. We cannot apply rational approaches anymore. Simon [3] pointed out that our rationality is bounded. If the number of data becomes too large, such problems as computational complexity emerges. Therefore, we have to solve the problem by trial and error.

But to control, we have to secure reproducibility. We have to expand rational approaches beyond the world of rationality. Then, how do we do that?

3. System Identification

System identification is crucial in system dynamics to establish a mathematical model. But let us discuss system identification in a much broader sense.

Let us first consider the problem how we can identify the name of a river. A river is continuously flowing and changing. So if we look at the flow, we cannot identify the name of the river. Therefore, we look around and we find something that does not change with time, such as trees, mountains, etc. They serve as identification points.

In the field of control, the same idea works. Let us take arc for example. There are many researches on arc, but we cannot predict its behavior. Yet, arc is used in welding and arc welding is used extensively. We cannot predict its behavior rationally, so we look around the molten pool and we find points which behave rationally. Thus, we can control arc based upon these points. The idea is the same as we identify the name of a river.

Engineering expanded its world beyond the world of rationality by identifying proper controllable points.

But if we consider bicycle riding, there are no such controllable points around. The situation is the same as in swimming. We are in the flow.

Then, how can we control not from the outside, but from the inside?

4. Adaptable Modeling

Thus, what is needed for human motion control is adaptable modeling. In system dynamics or software engineering, their concepts of adaptive modeling are how they can make their model adaptive to the changes. But in the case of human motion control, we do not have any models. We have to develop an *adaptable* model.

In other words, we need to develop a new approach which permits us to develop a model one after another in response to the changing situations. This is nothing other than learning. We learn to grow.

To be able to develop such an adaptable model, we must be situationally aware or fully cognizant of the situation.

Then, how can we develop such a growing model or an adaptable model?

5. Detection of emotion from face: Our previous research

This challenge brought us back to our previous research. We used to work on detecting emotion from face. We tried many image processing techniques, but without success. After many failures, we realized that everybody can easily detect emotion from cartoon face. It is very simple and in black and white. This led us to change our policy. We changed from detection to recognition. Cartoon face is a pattern. We understand how a cartoon character is feeling based on a pattern. We compare face patterns and if the pattern we are looking at matches with the pattern of one emotion, for example, happy, then we know the character is happy.

We simplified face images into patterns and compared the pattern of simplified face image with the cartoon face model pattern as shown in Figure 2. This approach was very successful. We do not have to spend much time for processing and in addition, our faces vary from person to person and even one man's face varies from time to time, but such wide variations can be processed without any difficulty [4],[5].

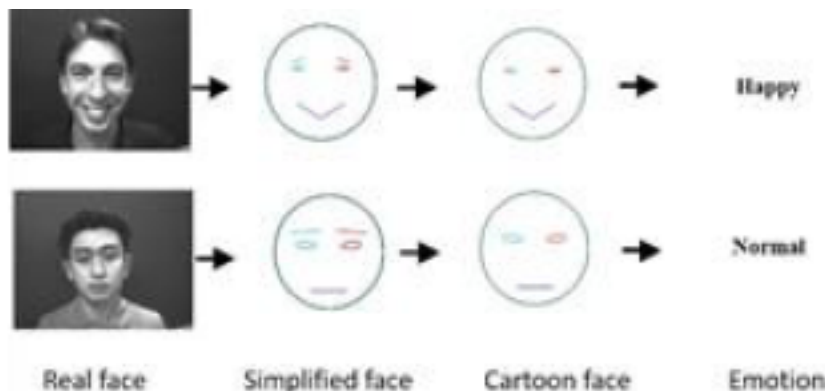


Figure 2. Detection of emotion from face using cartoon face model.

This experience taught us a lesson. We were trying to *detect* emotion, but if we try to *recognize* it, it is much easier. In other words, if we have a model of some kind, no matter how rough it may be, we can develop an adaptable model easily. Besides, if we introduce a pattern approach, we can process multi-dimensional information at once. In the case of other techniques, we have to apply it one after another.

Human motion is multi-dimensional, but if we introduce a pattern approach, we understand at once, if our motion is good or not. But how can we evaluate adaptability? In order to be adaptable, we have to know how much good or bad our motion is. If we can measure it, then, we can adapt our motion to the changing situation adequately.

6. Mahalanobis Taguchi System (MTS)

Mahalanobis Taguchi System or MTS [6],[7] is a very unique technique for pattern recognition. It introduces Mahalanobis Distance (MD), which reduces multi-dimensional information into one dimension and enables us to compare different patterns quantitatively with one single measure MD.

The basic idea of MTS is to define Unit Space and to compare a sample with this Unit Space using Mahalanobis Distance (MD). MD shows how much a sample pattern is different from the Unit Space (ideal pattern). A threshold MD is set up and if the MD of a sample pattern is smaller than this MD threshold, then a sample pattern is determined to belong to the Unit Space pattern (Figure 3).

For example, if we are going to recognize the image of a number, first we divide the image of numbers 0, 1, 2,---, 9 into pixels. We collect these unit space samples and average them and obtain Unit Space for 0, 1, 2,---9 (Figure 4).

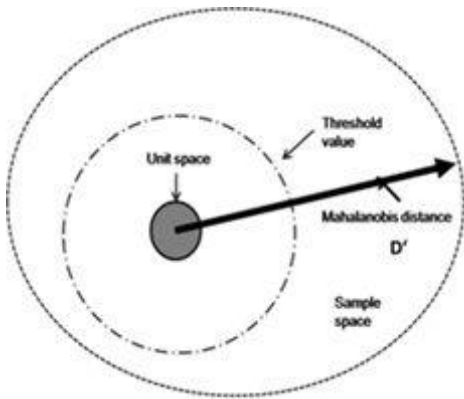


Figure 3. Mahalanobis Taguchi System.



Figure 4. How Unit Space is obtained.

7. Recognition Taguchi (RT) method

MTS developed from MT method to MTS method, and further to old MTA method, new MTA method, TS method, T method (1), T method (2) and then to RT method. RT method is the newest which was developed very recently. Recognition Taguchi (RT) method serves for the current purpose.

The procedures of RT method are as follows.

- (1) Determine feature items to define a pattern
- (2) Collect sample data to constitute Unit Space
- (3) Define Unit Space
- (4) Calculate Mean MD (MDmean) of Unit Space samples
- (5) Determine Threshold MD (MDthreshold) based on MDmean.
- (6) Calculate MDsample for a sample

When we come across the number image, we compare this image sample with these Unit Spaces and calculate their MDs. The smallest MD leads us to recognize what number it is.

The primary benefits of RT method are

- (1) We can prepare a large number of unit spaces.
- (2) They can be processed very rapidly.

For example, if we process numbers, we only need to prepare 10 (0 to 9) sample spaces. But in the case of body movements, we need to prepare a large number of sample spaces, but they can be accommodated and processed rapidly without any mathematical difficulty.

In fact, we can process 10s of thousands of unit sample spaces. This is very much different from other MTS methods. Most of them need to limit the number of unit sample spaces and their processing speed is slow.

It should be noted that original RT is developed for Recognition. Thus, it is named Recognition Taguchi. But, although the procedure is the same, we used it for cognition. So to be exact, the method used here should be called Cognitive Taguchi (CT) method.

8. Evolutionary Cognitive Taguchi (ECT) method

The regular RT method defines Unit Space after assembling many samples as shown in Figure 4. But to develop cognitive capabilities, we need to introduce a new way of defining Unit Space.

Instead of defining Unit Space after all repetitions, we update Unit Space after each success. After many successful repetitions, Unit Space will be the same as the one obtained by the regular procedures. But this process of updating provides us with information about how much we are improving and whether we are on the right track or not.

9. Intrinsic cognitive pattern

Most cognitive pattern approach are from outside. But what we are discussing here is a cognitive pattern approach from inside. It will satisfy our intrinsic motivation and our needs for growth, as described next.

Gallwey pointed out in his book “The Inner Game of Tennis” what an important role our bodies play in tennis [8]. Our muscles are different from person to person, so to win a game, there is no explicit way, but we must fully utilize our embodied cognition. We have to learn from failures. That is his message. Intrinsic cognitive pattern approach shares the same idea.

Brain science, neuroscience, etc. are progressing rapidly so intrinsic cognitive pattern or network is expected to be made clear soon, but at present, we have to rely on such technologies as Motion Capture, Eye Tracking, EMG, etc. But no matter what technology may be used, we can define intrinsic cognitive pattern based on these data as described above.

10. Self-Determination Theory (SDT)

Deci and Ryan proposed Self-Determination Theory (SDT) [9][10][11]. They pointed out that there are two kinds of motivations; extrinsic and intrinsic. Extrinsic motivation is in economics term, a reward.

We feel happier, even if the job is the same, if we are motivated internally and do it. Traditional engineering has been making efforts to satisfy our extrinsic motivations. We made efforts on how we produce better quality and better functioning products. Although voice of customers has been emphasized, it is only considered within the framework of external motivation. We would like to make decisions ourselves. That is why Deci and Ryan called their theory SDT.

Another important point they pointed out is that we need to grow. Learning is growing. That is why we feel happy when we learn to ride a bicycle. But most machines today deprive us of such opportunities of self-determination or learning to grow. We have to re-consider our design as will be discussed next.

Intrinsic cognitive pattern approach is expected to be one of the solutions.

11. Tolerance

By introducing this intrinsic cognitive pattern approach, we can satisfy intrinsic motivations of users. We can provide a wide margin for users to decide and act in their own way. If such a wide margin is secured in design, then it is expected that man and machine can work more happily together and it is expected we can reduce human errors drastically as we did with technical failures.

Such AI technology as Deep Learning will help us to set such a margin. But current AI needs computer power, but RT is fast enough and needs far less computer capability. Thus, it is much easier to apply.

12. Sense of balance

Physiology has made clear that we balance our body using our deep sensation, or proprioception, which controls our position and movements. It is important to note that proprioception varies widely from person to person, as does skin sensation. In fact, proprioception comes from Latin proprius, meaning “one’s own”, “individual”. So, it is natural that our movements differ from person to person.

We should remember that our sense of balance integrate all information. It processes multidimensional inputs from our sensors and sends signals to our actuators such as arms, legs, etc. Intrinsic cognitive pattern approach is deeply associated with our sense of balance.

13. Connected Machines

IoT is connecting machines rapidly. Traditional engineering made efforts to develop individual products with better quality and better performance. But now we need to design and operate machines as a team. So, just like a human body, we need a sense of balance or sense of coordination to run these machine teams.

Knute Rockne, American football coach, pointed out that a best team cannot be formed with 11 best players, but can be with players who play for the team adaptively.

Franz Beckenbauer, German soccer coach, introduced Libero system to make their team more adaptable to the changing situations on the pitch.

In other words, the importance of communication is rapidly increasing to be fully adaptive to the changing situations. Engineering systems need to be re-designed to communicate better not only between machine and machine, but also between man and machine with ample tolerance for human errors..

Intrinsic cognitive pattern approach described here is expected to be very effective and will play an important role in this age of communication.

14. Summary

An intrinsic cognitive pattern approach to motion control is proposed, using revised version of Recognition Taguchi method, which is one of Mahalanobis Taguchi Systems (MTS). The greatest feature of MTS is to reduce multi-dimensional informatio to one

dimension. Thus, it can compare patterns quantitatively using Mahalanobis distance (MD).

As the problem we are tackling here is not to recognize patterns, but to realize how we can be successful in our motion, RT method is reframed as Cognitive Taguchi method and it helps us to learn motion more effectively by comparing motion pattern one after another. MD provides us with a quantitative measure so we can understand how we should improve our motion to be successful.

The uniqueness of our approach is while most cognitive approaches look at a system from outside and identify the system parameters to control motion, our approach look at a system from inside and entrust decisions to humans.

This is because the number of degrees of freedom in human motion is too large, we take note of the superb human sense of balance. This enables us to allow a much wider margin in design for humans, which permits our trials and errors. Thus, it will satisfy our desire to make decisions ourselves and our needs to grow.

Although this approach is developed to control our motion, it will also help to respond to the change IoT is bringing about, i.e., connected machines. Traditional engineering has been paying attention to an individual machine, but what is facing us now is a team of machines. It is expected our approach will also help to respond to such requirements, too.

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